

European and Japanese Spacecraft Examines Mercury's Pole

A spacecraft recently sent back some of the best close-up pictures yet of the planet Mercury's north pole.

The European and Japanese robotic explorer went as close as 295 kilometers above Mercury's night side before passing directly over the planet's north pole.

The European Space Agency (ESA) released images that show craters with deep shadows on our solar system's smallest planet.

ESA noted that the edges, or rims, of the craters called Prokofiev, Kandinsky, Tolkien and Gordimer create permanent shadows. As a result, these unlit craters are some of the coldest places in the Solar System even though Mercury is the closest planet to the sun.

Cameras also took images of neighboring volcanic plains known as Borealis Planitia. ESA noted that these are "Mercury's largest expanse of 'smooth plains' and were formed by the widespread eruption of...lava 3.7 billion years ago."

Also appearing in the images released by ESA is Mercury's largest impact crater, the Caloris Basin, which covers more than 1,500 kilometers.

This image provided by European Space Agency shows close-up photos of Mercury's north pole taken by the European-Japanese spacecraft BepiColombo. (European Space Agency via AP)

This was the sixth and final flyby of Mercury for the BepiColombo spacecraft since its launch in 2018. The recent move put the spacecraft on a path to enter orbit around Mercury late next year. The spacecraft holds two orbiters, one from Europe and the other from Japan, that will circle the planet's poles.

In a statement on its website, ESA noted that Mercury, a small, dry planet, is "the least explored planet of the inner Solar System. Learning more about Mercury will shed light on the history of the entire Solar System."

The spacecraft is named for the late Giuseppe (Bepi) Colombo, an Italian mathematician. Colombo played a part in NASA's Mariner 10 mission to Mercury in the 1970s and, 20 years later, in the Italian Space Agency's special satellite project that flew on the U.S. Space Shuttle.

I'm John Russell.

Marcia Dunn reported this story for the Associated Press. John Russell adapted it for VOA Learning English.

Vocabulary

shadow - v. บังแสง, ทอดเงา (ก่อให้เกิดภาพมืดบนพื้นผิวเมื่อมีวัตถุบังแสง)

crater - n. หลุมอุกกาบาต, ปล่องภูเขาไฟ, แอ่ง (พื้นที่ต่ำหรือรอยบุ๋มที่เกิดจากการกระทบ)

expanse - n. พื้นที่กว้างใหญ่, บริเวณกว้างขวาง (ขอบเขตหรือระยะทางอันกว้างใหญ่ของสิ่งที่แผ่กระจายออกไป)

erupt - v. ปะทุ, ระเบิด (เช่น ภูเขาไฟ) (พุ่งออกมาหรือปล่อยออกมาอย่างกะทันหัน มักเป็นลาวาหรือไอน้ำที่อัดแน่นอยู่)

shed light - idiom. ให้ความกระจ่าง, ขยายความรู้, เผยข้อมูล (ขยายหรือเพิ่มพูนความรู้)

Reading Comprehension Quiz

1. What does the word 'Embed' in Paragraph 1 most likely suggest about the original article's format?

It is an instruction for printing the article.

It indicates the article is part of a larger book.

It signifies the presence of a non-textual media element, such as a video or audio.

It means the article's content is deeply integrated with scientific journals.

2. What specific discovery or event is highlighted in the opening sentence of the article's content?

The launch of a new European satellite.

The first human mission to Mercury's north pole.

The recent return of high-quality close-up pictures of Mercury's north pole by a spacecraft.

A detailed analysis of Mercury's south pole climate.

3. What was the closest distance the robotic explorer achieved above Mercury's night side?

2,950 kilometers

295 kilometers

29.5 kilometers

2,950 miles

4. Who released the images showing craters with deep shadows on Mercury?

NASA

The Japanese Space Agency

The European Space Agency (ESA)

The Associated Press

5. What characteristic of certain craters on Mercury causes them to be among the coldest places in the Solar System?

Their location at the south pole.

Their extreme depth and volcanic activity.

Their unlit areas caused by permanent shadows created by crater rims.

Their composition of highly reflective ice.

6. Which of the following is NOT mentioned as a name of a crater creating permanent shadows on Mercury?

Prokofiev

Kandinsky

Galileo

Tolkien

7. What are Borealis Planitia described as in the article?

Mountain ranges near Mercury's equator.

Ancient riverbeds on Mercury's surface.

Mercury's largest expanse of 'smooth plains' formed by widespread lava eruption.

Impact basins filled with water ice.

8. When were the Borealis Planitia plains formed?

Approximately 3.7 million years ago.

Approximately 3.7 billion years ago.

During Mercury's initial formation.

Recently, due to ongoing volcanic activity.

9. What is the name and approximate size of Mercury's largest impact crater mentioned in the images?

Prokofiev, covering 500 kilometers.

Caloris Basin, covering more than 1,500 kilometers.

Borealis Planitia, spanning 3,700 kilometers.

Kandinsky, about 295 kilometers wide.

10. According to the image caption, which agency provided the close-up photos of Mercury's north pole?

The Japanese Space Agency

NASA

11. What is the total number of flybys completed by the BepiColombo spacecraft, and when was it launched?

Five flybys since its launch in 2010.

Six flybys since its launch in 2018.

Ten flybys since its launch in 2015.

Two flybys since its launch in 2020.

12. What is the next major phase planned for the BepiColombo spacecraft?

To return to Earth for maintenance.

To perform a final flyby of the Sun.

To enter orbit around Mercury late next year.

To send its two orbiters to Mars.

13. Why does ESA state that learning more about Mercury is important?

To understand the planet's potential for future human colonization.

Because it is the most explored planet in the inner Solar System.

To shed light on the history of the entire Solar System.

To find new sources of water and minerals.

14. Who is the BepiColombo spacecraft named after?

An Italian astronomer named Bepi Galileo.

A Japanese engineer who designed the spacecraft.

The late Giuseppe (Bepi) Colombo, an Italian mathematician.

A famous European space explorer from the 20th century.

15. What was one of Giuseppe Colombo's key contributions to space exploration?

He designed the BepiColombo spacecraft's propulsion system.

He played a part in NASA's Mariner 10 mission to Mercury in the 1970s.

He discovered new craters on Mercury's north pole.

He was the first astronaut to orbit Mercury.